Measuring Microbial Community Dynamics using Physiology and Genomics

Jessica R. Bernardin, M.Sc.
Advisor: Leonora Bittleston, Ph.D

BOISE STATE UNIVERSITY

BACKGROUND

Ecosystems are built on the foundation of complex relationships between communities of organisms and their environment. Studying these communities is challenging due to the vast geographic ranges of many organisms and the variability present in natural environments (Brown 1999).

Microbial communities provide a good tool to measure patterns in community structure and function. Understanding how microbial communities function, how those functions impact their host, and predicting the functional capabilities of complex communities will bolster inferences made on observed patterns across successional and spatial gradients.

I will measure the impact of changing biotic and abiotic conditions on the microbial function and composition in the *Sarracenia purpurea* model system, along with downstream impacts on host health. Further, I will identify general patterns in microbiome composition and function across an altitudinal gradient.

RESEARCH QUESTIONS

1. How does the general function of a microbial community impact the associated host's health and nutrient acquisition?
2. How do environmental shifts impact host-associated microbiome composition and function?
3. What are the general trends in microbiome functional and compositional succession and are those patterns impacted by changes in altitude?
4. How resilient are the host-associated microbial communities of *S. purpurea* to shifts in prey type and what effect does that have on microbial recruitment and function?

METHODS

- EcoPlates were used for physiological profiling of microbial communities (OD_{590}) over 4 days.
- Chitinase and protease assays used LAMC and 4-MU chitobioside substrates and measured as fluorescence on plate reader.
- Growth metrics measured as OD_{590} over three days in R2A medium and calculated using growthcurveR package in R.
- Other physiological metrics conducted but not presented include microbial community respiration (MicroResp), biomass using fluorescent microscopy (BacLight Live/Dead stain), plant morphometric measurements, and plant elemental analysis (C:N, Flash Inc.).

Does the function of the microbiome impact host health?

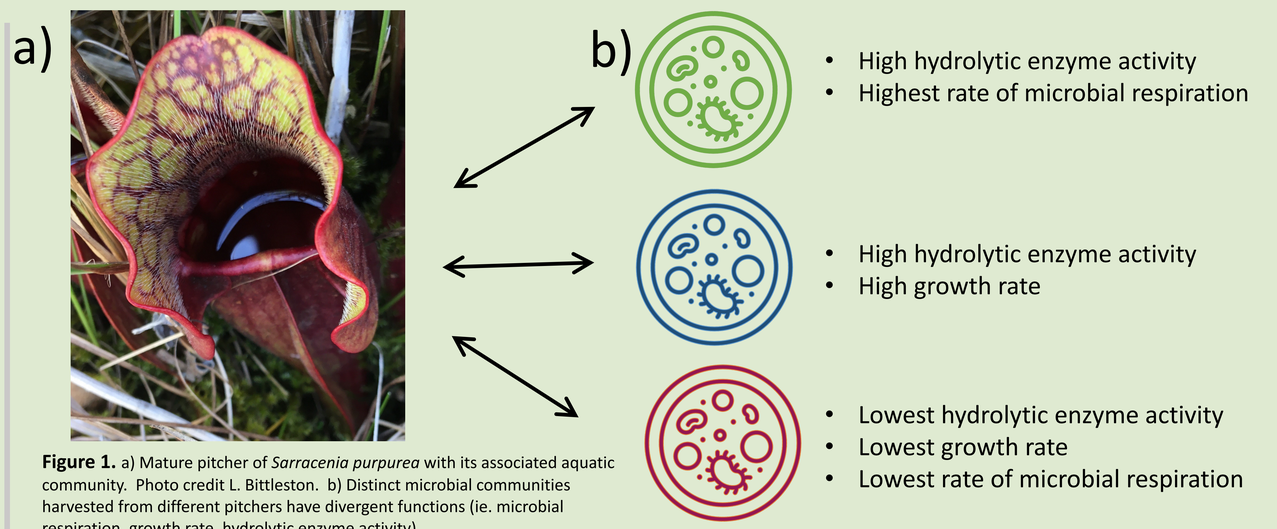

Figure 1. a) Mature pitcher of *Sarracenia purpurea* with its associated aquatic community. Photo credit L. Bittleston. b) Distinct microbial communities harvested from different pitchers have divergent functions (ie. microbial respiration, growth rate, hydrolytic enzyme activity).

Divergent function based on substrate use

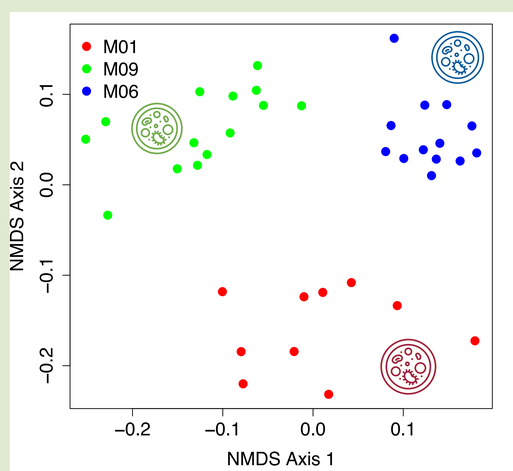

Figure 2. Non-metric multidimensional scaling (NMDS) plot based on a Bray-Curtis dissimilarity matrix from Biolog Ecoplate physiological profiles for three pitcher plant microbial communities grown *in planta*. Microbial communities are coded M01 (red), M06 (blue), M09 (green), each point represents the microbial community in one individual pitcher. These communities show divergence in substrate use during the first week of the experiment.

A non-significant effect on host photosynthesis

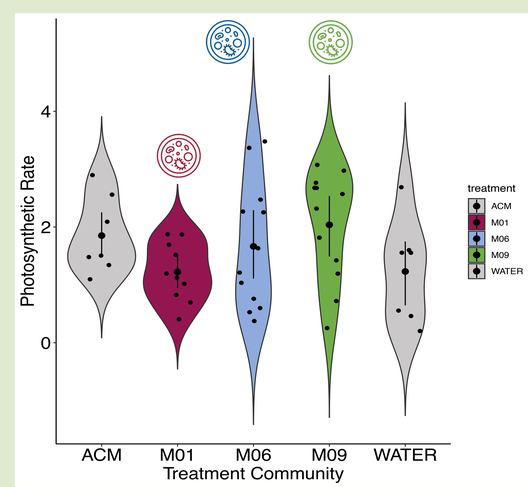

Figure 3. Photosynthetic rate (µmol CO_2 m^{-2} s^{-1}) for 43 plants after 8 weeks of hosting one of three microbial community treatments. Mean photosynthetic rate shows non-significant differences between the treatment groups and the controls, water and media controls.

CONCLUSIONS

- Microbial communities from wild pitcher plants show different functions when comparing respiration, growth rate, and hydrolytic enzyme activity (Figure 1b).
- Microbial communities show clear differences in function (Figure 2), given a panel of 31 different substrates, microbial communities utilize them differently and these differences were stable over time in culture.
- When experimentally introduced to host plants, microbial community function had a non-significant effect on host photosynthetic efficiency (Figure 3).
- Microbial community function shifts through time within the plant (data not shown).
- Results on the effect of microbial community function on host nutrient acquisition are forthcoming.

PROPOSED FUTURE WORK

- Metagenomic and metatranscriptomic analysis of microbial community samples from growth chamber experiment
- Factorial experiment conducted in growth chambers testing the effect of environmental stress on community function
- Swiss field experiment measuring compositional and functional changes across time and altitude
- Swiss field and common garden experiments measuring the microbial functional response to changes in prey

Acknowledgements:
- Dr. Leonora Bittleston and all the members Bittleston Lab
- Dr. Marie-Anne de Graaff and Lola Klamm
- Dr. Sarah Gray and Dr. Kevin Feris
- Dr. Marcelo Serpe and Teo Geisler
- Dr. Jen Pierce, Bill Freutel, and the Earth, Wind, and Fire Lab
- Dr. Linda Reynard
- Dr. Erica Young (UW) and other members of the MTM group

References:

1. Brown JH. 1999. Macroecology: Progress and Prospect. Oikos. 87(1):3–14.